



FREE RESOURCE FOR SCHOOL TEAMS

School FA Starter Kit



Five functional analysis formats for schools: step-by-step task analyses, printable datasheets, and graphing pages.

from **Robert Spain, BCBA, IBA** · Behavior School

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Functional Behavior Assessment in a School Setting · CalABA BehaviorLive

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What's inside

How to use this kit (and a safety note)	3
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Part 1 · Task analyses

Analog / formal FA	4
Synthesized FA (PFA / IISCA)	5
Trial-based FA	6
Latency-based FA	7
Precursor FA	8

Part 2 · Printable datasheets

Standard (Analog) FA	9
Practical Functional Assessment (PFA / IISCA)	10
Trial-Based FA (TBFA)	11
Latency-Based FA	12
Precursor FA	13

Part 3 · Graphing

The FA graphing template (with an example graph)	14
Blank graph page: Standard (analog) FA	15
Blank graph page: PFA / IISCA	16
Blank graph page: Trial-based FA	17
Blank graph page: Latency-based FA	18
Blank graph page: Precursor FA	19

Part 4 · Reading and next steps

Key references	20
Where to go from here	21

How to use this kit

Thanks for spending an hour with me on school-based FA. This kit is the paper side of that talk: the five FA formats we walked through, written as steps, plus the datasheets and graph pages I use to run them in schools.

Assess, don't guess.

1. **Start with a testable hypothesis.** Use your FBA (interview, records, observation) to say what you think is maintaining the behavior. If no one can test it, it's still a guess.
2. **Pick the format that fits the student and the setting.** Trial-based fits inside classroom routines. Latency and precursor formats keep exposure low when behavior is dangerous. Synthesized (IISCA) tests what the team describes. Analog is the specialist-led, controlled version.
3. **Read the steps, then print the matching datasheet and graph page.** Each task analysis page tells you which pages go with it.
4. **Graph as you go and decide as a team.** Look for clear differences between test and control before anyone writes the BIP.
5. **Link the function to the plan:** prevention, a replacement skill to teach, and reinforcement for it.

Safety first: before you run any FA

- **Trained staff only.** A qualified behavior analyst should design and supervise the FA. Everyone running sessions is trained on the procedure first.
- **Consent first.** Get team agreement and guardian consent as your district's policy requires before any session.
- **Stop criteria in writing.** Decide ahead of time what ends a session (injury risk, severity, distress, withdrawal of assent), and stop right away when you see it.
- **This kit is not a substitute for supervision,** training, or clinical judgment. It's a starting point for teams that already have those in place.

Keep completed datasheets confidential under FERPA and your district's policy. Use a student ID or code on anything that leaves the file.

And when someone asks what's driving the behavior, it's okay to say: "I don't have a magical quick answer, but I have a process."

— Robert Spain, BCBA, IBA

Analog / formal FA: task analysis

- 1 Confirm specialist-led need, risk screen, stop criteria, assent/withdrawal, and supervision before any session.
- 2 Operationally define the target response class; set session length (commonly 5–10 min) and measurement (rate or interval).
- 3 Prepare a controlled setting and condition materials (attention scripts, demand tasks, tangibles if used, ignore/alone arrangement, play/control materials).
- 4 Play / control: preferred leisure, noncontingent attention as designed, no demands; deliver no differential consequence for target (or per protocol).
- 5 Attention: low/diverted attention; contingent attention (reprimand/concern) contingent on target.
- 6 Escape / demand: present academic or instructional demands; contingent break / demand removal on target (prompting rules per protocol).
- 7 Alone or ignore: minimal stimulation / no social consequences for target (choose alone vs ignore based on safety and self-injury risk).
- 8 Tangible (if hypothesized): restrict preferred item; contingent access on target.
- 9 Run conditions in a multielement (alternating) design with integrity checks; honor stop criteria immediately.
- 10 Graph rate/intervals by condition; require differentiation + replication across sessions.
- 11 Interpret: elevated responding in one (or more) test condition(s) vs play/control supports function(s); undifferentiated → redesign, extend, or choose adapted FA.
- 12 Link confirmed function(s) to BIP components; document school feasibility limits of pull-out analog testing.

Primary citation: Iwata, Dorsey, Slifer, Bauman, & Richman (1994/1982); optional: Saini et al. (2020)
Datasheet: page 9 · Blank graph: page 15 · Full references: page 20

Synthesized FA (PFA / IISCA): task analysis

- 1 Conduct an open-ended interview with the teaching team (perfect day, rapid escalation, what happens when work/preferred access changes).
- 2 Synthesize the evocative context and reinforcer class from interview (+ brief observation if needed); do not isolate variables by default.
- 3 Define target and safe precursors; set phase length (often 3–5 min), timer, datasheet, and stop/assent rules.
- 4 Materials: synthesized reinforcers for HRE/control + evocative demands/item removal for test.
- 5 Control (HRE): establish happy, relaxed, engaged — unrestricted access to synthesized reinforcers, zero evocative demands; target near zero.
- 6 Test: present the synthesized EO (e.g., remove preferred access + present demand).
- 7 On first precursor or target, immediately restore synthesized reinforcers / terminate EO (safety + dignity).
- 8 Alternate Control ↔ Test until clear differential responding with low exposure.
- 9 Record target, precursors, engagement, integrity, and safety stops each phase.
- 10 Decision: test > control differentiation confirms the synthesized contingency class.
- 11 Branch to skill-based treatment / function-based BIP using the same reinforcer class (FCR → tolerance → cooperation as appropriate).
- 12 If undifferentiated or unsafe: revise interview synthesis, tighten precursor definition, or select a different FA format.

Primary citation: Hanley et al. (2014); Rajaraman et al. (2022); Jessel et al. (2024)

Datasheet: page 10 · Blank graph: page 16 · Full references: page 20

Trial-based FA: task analysis

- 1 From FBA, pick hypothesized function(s) (attention, escape/demand, tangible) and the natural routines that hold those EOs.
- 2 Train implementers; use a stopwatch (not a countdown timer) for 1–2 min segments; print binary occurrence datasheet.
- 3 Wait for a natural opportunity — do not invent a disruption to start a trial.
- 4 Control segment (1–2 min): EO absent (e.g., no demand / free attention as designed); start stopwatch.
- 5 If target occurs in Control: record occurrence, abort that trial (do not run Test for that trial).
- 6 Test segment (1–2 min): immediately after a clean Control, introduce the EO; restart stopwatch.
- 7 If target occurs in Test: deliver the planned consequence, mark occurrence, end the trial.
- 8 If no occurrence by segment end: mark non-occurrence; end segment.
- 9 Collect multiple paired trials per hypothesized function across routines; sample enough for % occurrence comparison.
- 10 Graph percent of trials with occurrence — Test vs Control by function.
- 11 Decision: higher % in Test than Control supports that function; mixed/spoiled trials → revise EO/routine match.
- 12 Build routine-specific BIP (prevention, teaching, reinforcement) from supported function(s).

Primary citation: Bloom, Iwata, Fritz, Roscoe, & Carreau (2011)

Datasheet: page 11 · Blank graph: page 17 · Full references: page 20

Latency-based FA: task analysis

- 1 Confirm that repeated rate-based evocation is unsafe or inefficient; set risk screen and de-escalation plan.
- 2 Define target (and optional precursors); set max session time (e.g., 5:00) and no-response coding rule.
- 3 Prepare condition EOs (attention, demand, tangible, alone/ignore, play/control as needed) and latency log (MM:SS).
- 4 Present the condition antecedent / EO exactly as scripted.
- 5 Start stopwatch at the exact moment the EO is presented.
- 6 Do not react to non-target behavior unless it is a defined precursor or stop-rule event.
- 7 At the first target response: stop timing, record latency, terminate session and de-escalate.
- 8 If no response by max: stop at cap; record max (or “no response” per team rule).
- 9 Replicate across conditions in a multielement (or rapid alternating) arrangement.
- 10 Graph latency by condition (shorter latency = stronger control by that condition).
- 11 Decision: reliably shorter latencies in one test condition vs control supports function; similar latencies → redesign.
- 12 Select safer confirmed function for BIP; prefer adapted formats if analog-like condition set is still too exposing.

Primary citation: Thomason-Sassi, Iwata, Neidert, & Roscoe (2011)

Datasheet: page 12 · Blank graph: page 18 · Full references: page 20

Precursor FA: task analysis

- 1 Use descriptive FBA to list the severe target and earlier responses that reliably precede it.
- 2 Verify precursor–severe linkage (temporal reliability) before substituting precursor as the FA endpoint.
- 3 Operationally define the earliest safe precursor; write stop rules if severe behavior still emerges.
- 4 Arrange test vs control (or multielement) conditions from the hypothesized EO — same logic as standard FA, but consequence/termination at precursor.
- 5 Optional: map private events / avoidance with an ACT Matrix for verbal students (school team practice).
- 6 Present Test EO; start session timer (often 5–10 min).
- 7 On first defined precursor: deliver planned consequence / remove EO; do not wait for dangerous behavior.
- 8 In Control: EO absent; precursor should be low if conditions are well matched.
- 9 Record precursor by condition; separately log any breakthrough severe events and safety stops.
- 10 Graph precursor differentiation; treat breakthrough severe data as safety/integrity signals.
- 11 Decision: clear precursor differentiation + documented linkage → infer function for BIP; weak linkage → do not substitute.
- 12 Build BIP for inferred function (prevention, communication/skills, reinforcement); reassess if precursors drift or severe behavior continues.

Primary citation: Smith & Churchill (2002). Optional school note: ACT Matrix mapping (team tool — not required by 2002 method).

Datasheet: page 13 · Blank graph: page 19 · Full references: page 20

Standard (Analog) FA — Data Sheet

Continuous tally during fixed-length sessions (e.g., 5 or 10 min). Enter totals into Graphing Sheet → Analog FA tab.

Student name / ID: _____ Date: _____

BCBA / observer: _____ Session length (min): _____

Target behavior (operational definition): _____ Setting / materials: _____

How to use

- Mark the condition for each session. Tally every instance of the target during the session.
- Write the session total in the Total column. Graph responses per minute = Total ÷ session minutes.
- Alternate conditions (Attention, Demand/Escape, Play/Control, Tangible, Alone if used).

Session log

Date	Sess #	Start	Condition (circle one)	Behavior tally (hash marks)	Total
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		
			A D P T L		

Conditions: Attention | Demand (Escape) | Play (Control) | Tangible | Alone (if indicated). Print one sheet per student / day as needed.

Condition key (circle in Condition column): A = Attention D = Demand/Escape P = Play T = Tangible L = Alone

Practical Functional Assessment (PFA / IISCA) — Data Sheet

Synthesized Control (HRE) vs Test. Frame: Hanley et al. (2014). Typical phase 3–5 min. Graphing Sheet
→ Synthesized (IISCA) tab.

Student name / ID: _____ Date: _____

BCBA / observer: _____ Phase length (min): _____

Target / precursor definitions: _____ Synthesized EO (from interview): _____

Synthesized reinforcers (HRE): _____ Who implements: _____

How to use

- Control (HRE): continuous access to synthesized reinforcers; no evocative events.
- Test: present synthesized EO; at first precursor or target, restore HRE (end demand / return reinforcers).
- Tally targets and precursors separately. Do not wait for dangerous escalation.

Session log

Date	Sess #	Phase (C/T)	Start	End	Target tally	Precursor tally	Notes
		C T					
		C T					
		C T					
		C T					
		C T					
		C T					
		C T					

Phase: C = HRE Control T = Test. On Test: first precursor or target → restore HRE immediately.

Trial-Based FA (TBFA) — Data Sheet

Binary occurrence in Control then Test (about 1–2 min each). Bloom et al. (2011). Graphing Sheet → Trial-based FA tab.

Student name / ID: _____ Date: _____

BCBA / observer: _____ Target behavior: _____

Classroom / activity context: _____ Trial length (min): _____

How to use

- Run Control segment first, then Test for the same condition (Attention, Demand, or Tangible).
- If the target occurs in Control, abort that trial (do not run Test). Mark Control only.
- Mark Occurred / Did not for each segment. Later graph % of trials with occurrence by condition.

Trial log

Time	Trial #	Activity	Cond (A/D/T)	Control	Test	Abort?
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N
			A D T	☐ Occ ☐ No	☐ Occ ☐ No	Y N

Cond: A = Attention D = Demand T = Tangible. Abort = Yes if behavior in Control (skip Test).

Latency-Based FA — Data Sheet

Time to first response. Thomason-Sassi et al. (2011), JABA. End session at first occurrence. Graphing Sheet → Latency FA tab.

Student name / ID: _____ Date: _____

BCBA / observer: _____ Target behavior: _____

Max session time (e.g., 5:00): _____ No-response rule (circle): blank / enter max seconds: _____

How to use

- Start stopwatch when the condition EO is presented.
- Stop at the first instance of the target. Record MM:SS and convert to seconds.
- If no response by max time: follow the circled rule (leave blank OR enter max seconds).
- Shorter latency under a condition = faster onset under that EO.

Session log

Date	Sess #	Condition (circle)	Occurred?	Time to first (MM:SS)	Seconds	Notes
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			
		A D P T L	Y N			

Condition: Attention | Demand | Play (control) | Tangible | Alone. Occurred: Y / N. Graph seconds (lower = faster onset).

Condition key: A = Attention D = Demand P = Play (control) T = Tangible L = Alone

Precursor FA — Data Sheet

Reinforce / terminate at safe precursors. Smith & Churchill (2002). Optional ACT notes are team practice only. Graphing Sheet → Precursor FA tab.

Student name / ID: _____ Date: _____

BCBA / observer: _____ Session length (min): _____

Severe target (do not evoke repeatedly): _____ Precursor definition (safe earlier response): _____

EO / demand used in Test: _____ Safe consequence for precursor: _____

How to use

- Control: EO absent / reinforcers available as designed for this analysis.
- Test: present EO. At first precursor, remove EO / deliver safe consequence. Do not wait for severe target.
- Tally precursors (primary). Record target only if it still occurs. Optional Away/Toward notes are not part of the 2002 method.

Session log

Date	Sess #	Phase (C/T)	Precursor tally	Target tally	Away note	Toward note	Notes
		C T					
		C T					
		C T					
		C T					
		C T					
		C T					
		C T					
		C T					

Phase: C = Control T = Test. Primary DV = precursors. Keep dangerous PB near zero by ending at the precursor.

The FA graphing template

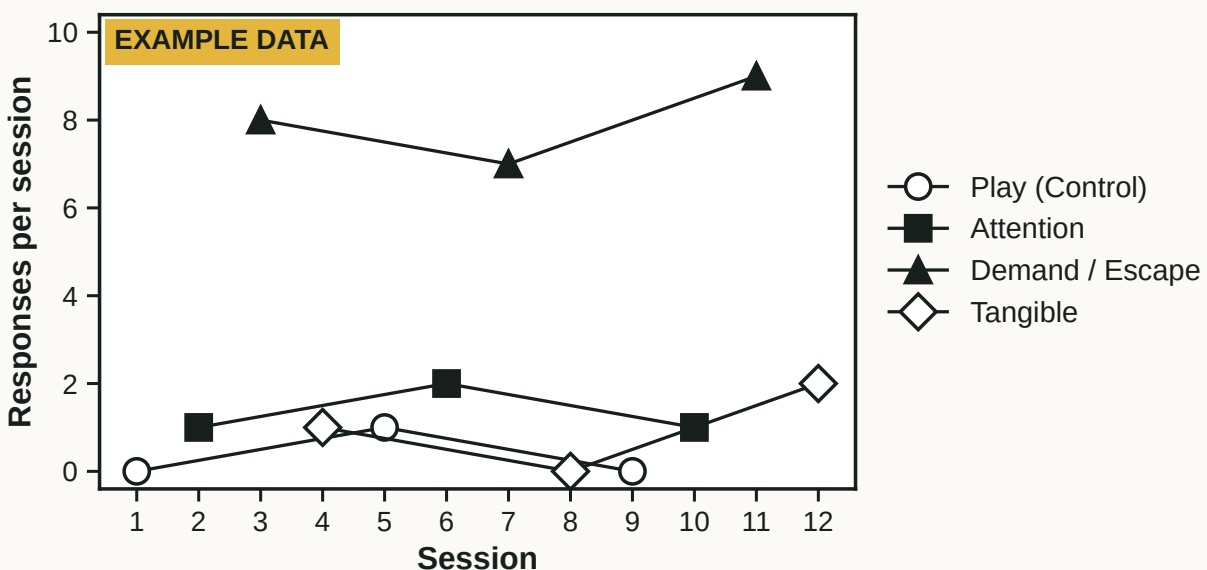
Hand-graphing works fine, and Part 3 gives you a blank graph page for each format. If you'd rather type, the **School FA Graphing Template** is a Google Sheet that draws the graph for you. **A copy link comes with your signup** at behaviorschool.com/fba. Make a copy for each student, and start on the Instructions tab.

The flow: collect on the datasheet → type the numbers into the white cells of the matching tab → the chart builds itself. Gray columns calculate on their own; blank rows are skipped, never plotted as zero.

Tab	What you type → what the chart shows
Analog FA	Session, condition, count, minutes → responses per minute by condition
Synthesized (IISCA)	Session, Control/Test, count, minutes → target per minute, Control vs Test
Trial-based FA	Block, condition, Y/N in control and test → % of trials per block
Latency FA	Session, condition, min:sec to first response → seconds (300 s cap)
Precursor FA	Session, Control/Test, precursor count, minutes → precursors per minute
EXAMPLE - Analog FA	Made-up demo data with a filled chart

Each condition gets its own color and marker shape, and a line joins only that condition's points, so data are never connected across conditions. Latency: pick your no-response rule (blank or the max) on the tab and write it down.

Example graph: Standard (analog) FA, multielement



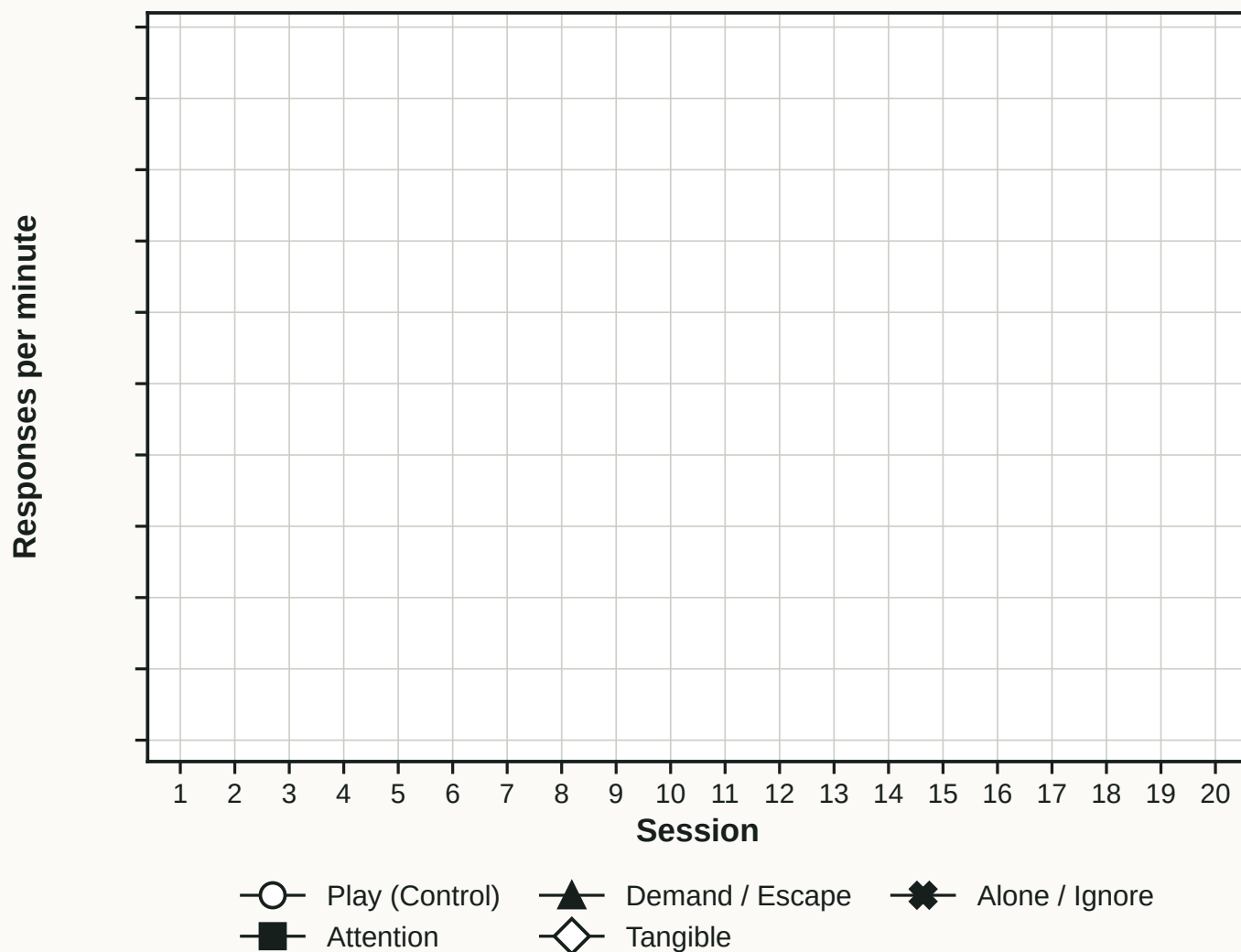
Example data. Made-up demo values from the template, not a real student. Here responding is highest in Demand and stays low in Play (control), which would support an escape function. Markers differ by condition, and lines connect points within each condition only.

Standard (analog) FA graph

Student ID / code: _____ Target behavior: _____

Dates: _____ Session / phase length: _____

Responses per minute by session, one marker shape per condition. Plot each session above its session number; connect points within a condition only. Write your y-axis scale on the tick marks.



Notes / decision:

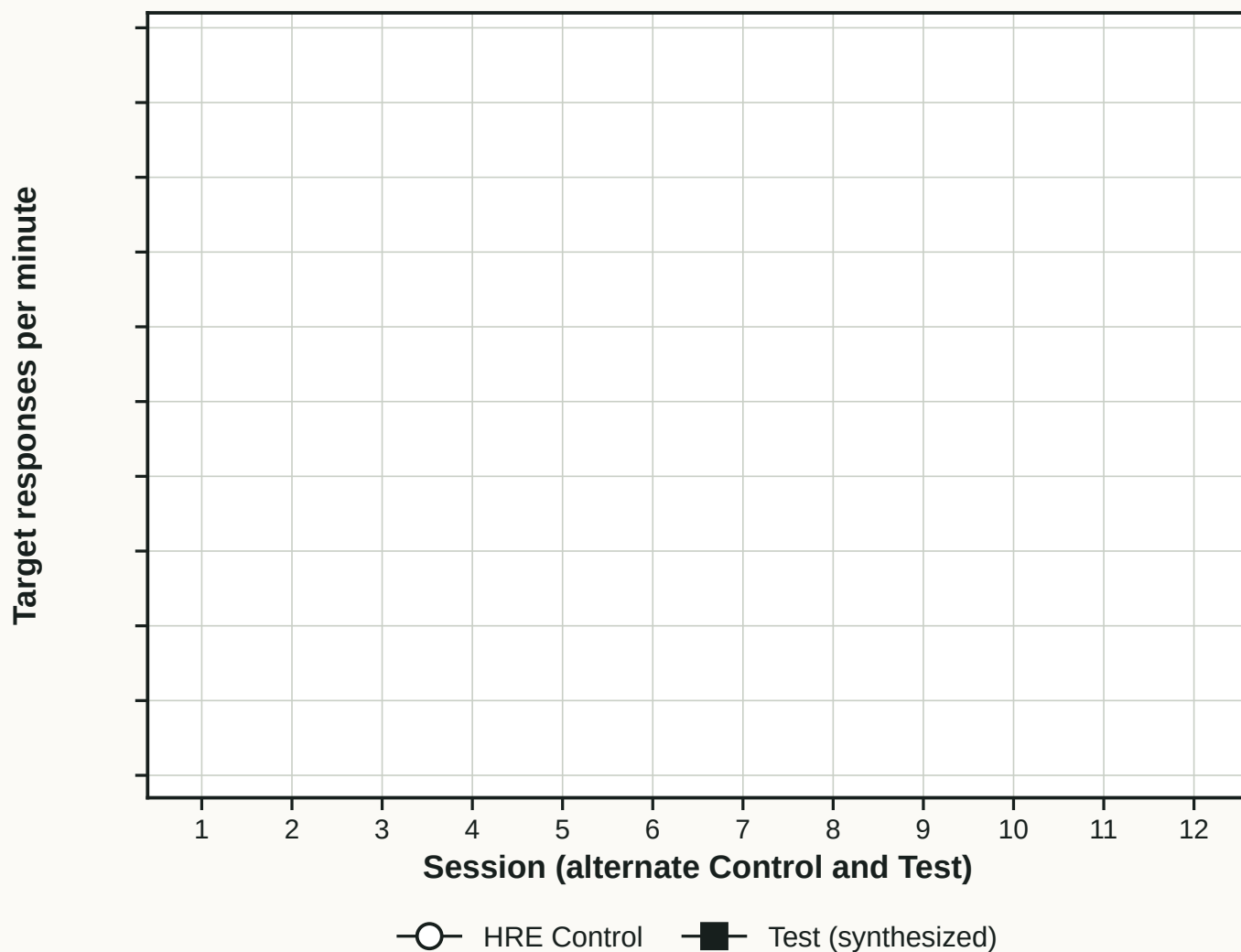
Data from the datasheet on page 9 · Sheet tab: Analog FA in the School FA Graphing Template (Google Sheet)

PFA / IISCA graph

Student ID / code: _____ Target behavior: _____

Dates: _____ Session / phase length: _____

Target responses per minute in each phase. Alternate HRE Control and Test; connect Control to Control and Test to Test. Write your y-axis scale on the tick marks.



Notes / decision:

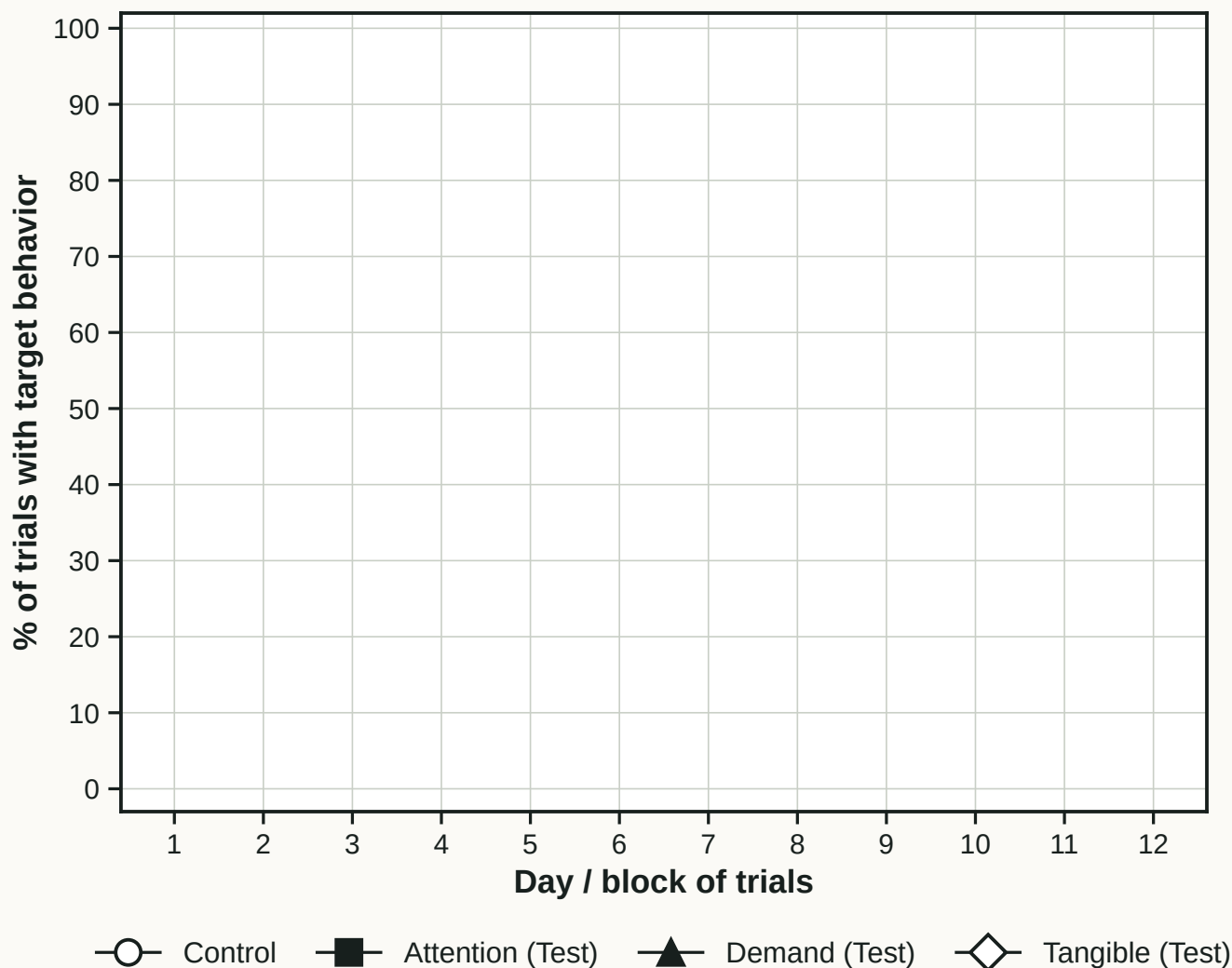
Data from the datasheet on page 10 · Sheet tab: Synthesized (IISCA) in the School FA Graphing Template (Google Sheet)

Trial-based FA graph

Student ID / code: _____ Target behavior: _____

Dates: _____ Session / phase length: _____

Percent of trials with the target behavior, by day or block. Control vs each Test condition; % = trials with occurrence ÷ trials run × 100.



Notes / decision:

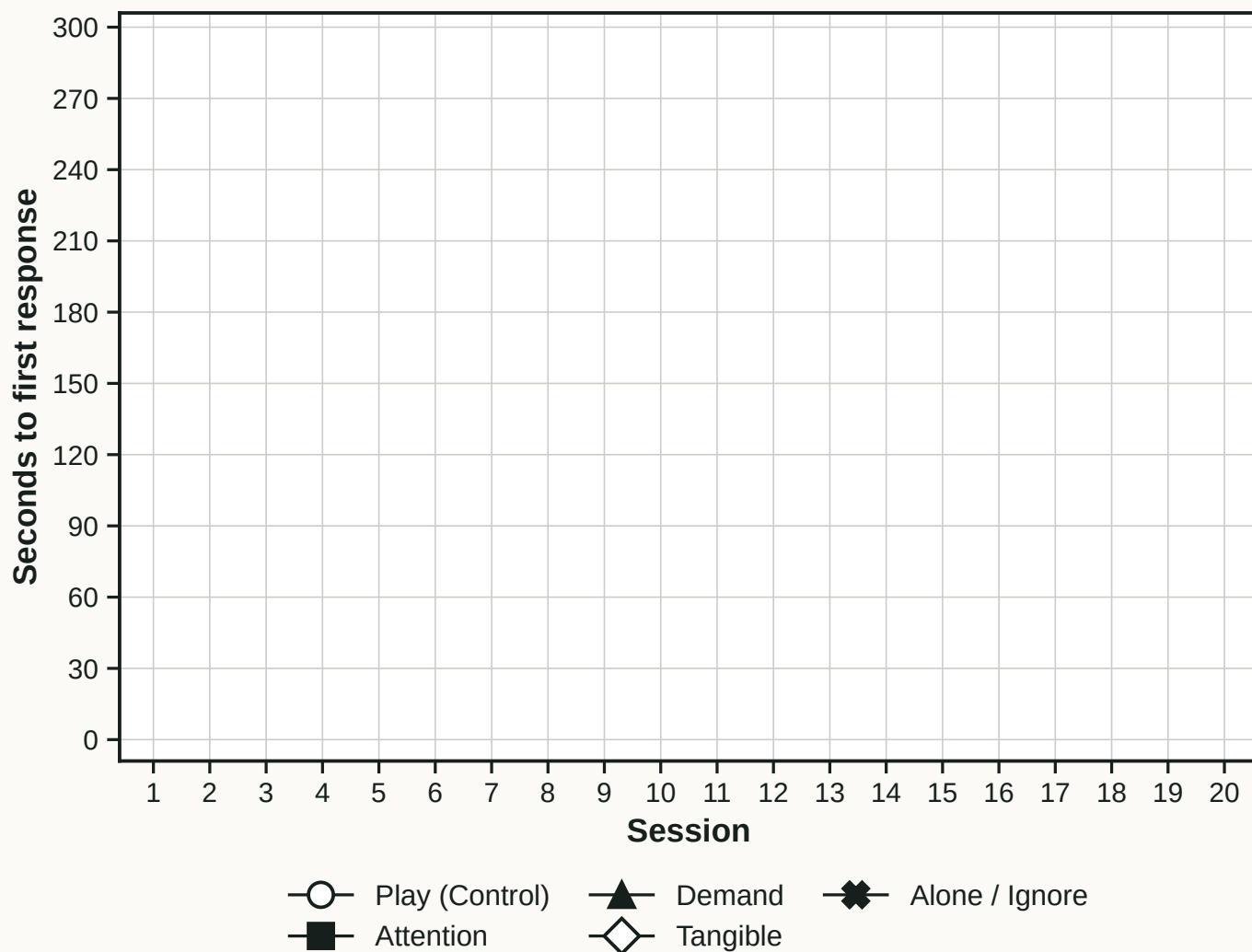
Data from the datasheet on page 11 · Sheet tab: Trial-based FA in the School FA Graphing Template (Google Sheet)

Latency-based FA graph

Student ID / code: _____ Target behavior: _____

Dates: _____ Session / phase length: _____

Seconds to the first response, by session and condition. Default scale is a 5:00 (300 s) cap; mark which no-response rule you used.



Notes / decision:

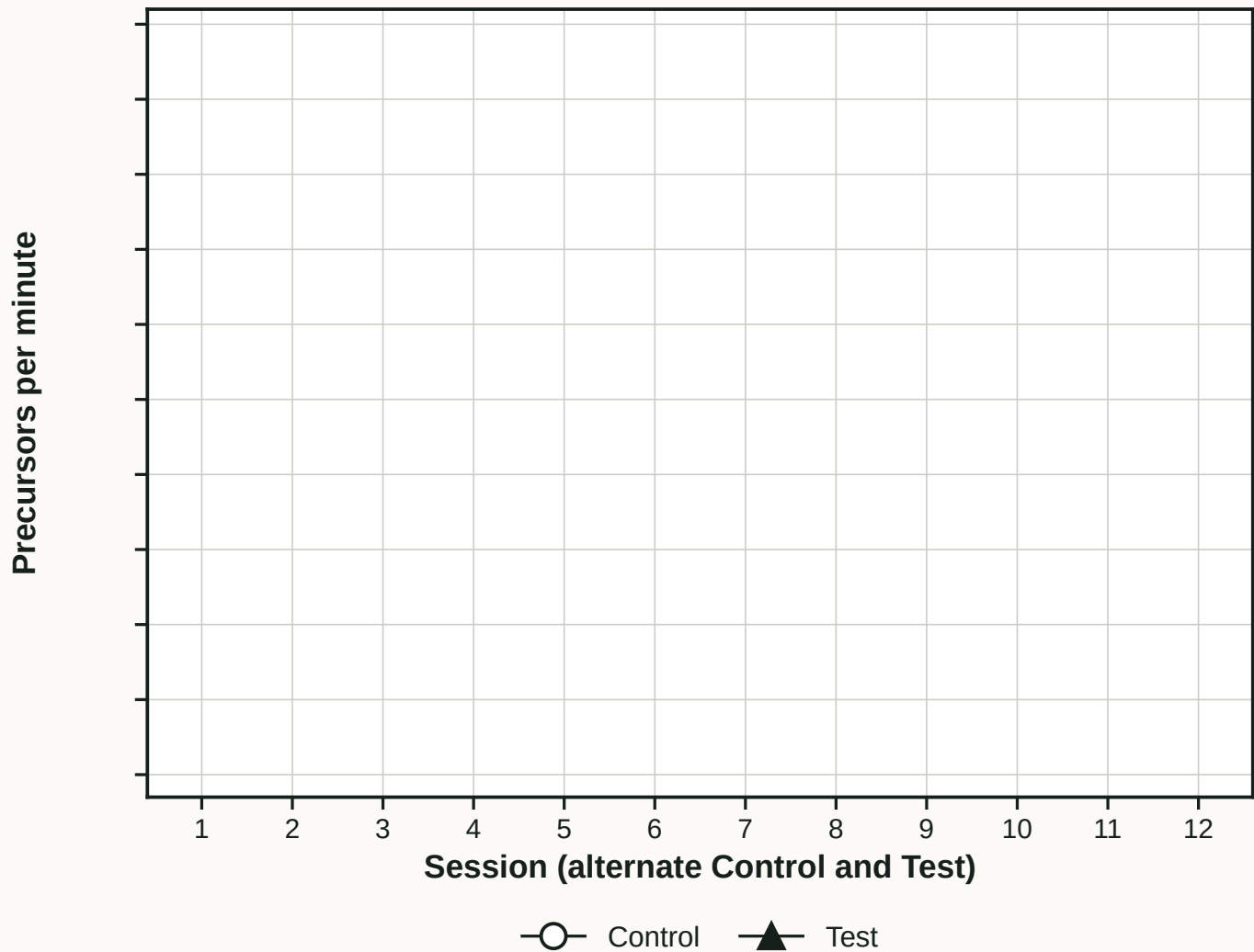
Data from the datasheet on page 12 · Sheet tab: Latency FA in the School FA Graphing Template (Google Sheet)

Precursor FA graph

Student ID / code: _____ Target behavior: _____

Dates: _____ Session / phase length: _____

Precursors per minute, Control vs Test. Log any breakthrough severe events separately, in the notes, as safety/integrity signals. Write your y-axis scale on the tick marks.



Notes / decision:

Data from the datasheet on page 13 · Sheet tab: Precursor FA in the School FA Graphing Template (Google Sheet)

Key references

The primary sources behind the five task analyses, as listed in the webinar's reference slides.

Bloom, S. E., Iwata, B. A., Fritz, J. N., Roscoe, E. M., & Carreau, A. B. (2011). Classroom application of a trial-based functional analysis. *Journal of Applied Behavior Analysis*.
<https://doi.org/10.1901/jaba.2011.44-19>

Hanley, G. P., Jin, C. S., Vanselow, N. R., & Hanratty, L. A. (2014). Producing meaningful improvements in problem behavior of children with autism via synthesized analyses and treatments. *Journal of Applied Behavior Analysis*. <https://doi.org/10.1002/jaba.106>

Iwata, B. A., Dorsey, M. F., Slifer, K. J., Bauman, K. E., & Richman, G. S. (1994). Toward a functional analysis of self-injury. *Journal of Applied Behavior Analysis*, 27(2), 197–209. (Reprint of 1982 *Analysis and Intervention in Developmental Disabilities*.)

Jessel, J., Fruchtmann, T., Raghunauth-Zaman, N., Leyman, A., Lemos, F. M., Val, H. C., Howard, M., & Hanley, G. P. (2024). A Two Step Validation of the Performance-Based IISCA: A Trauma-Informed Functional Analysis Model. *Behavior Analysis in Practice*.
<https://doi.org/10.1007/s40617-023-00792-2>

Rajaraman, A., Hanley, G. P., Gover, H. C., Ruppel, K. W., & Landa, R. K. (2022). On the Reliability and Treatment Utility of the Practical Functional Assessment Process. *Behavior Analysis in Practice*. <https://doi.org/10.1007/s40617-021-00665-6>

Saini, V., Fisher, W. W., Retzlaff, B. J., & Keevy, M. (2020). Efficiency in functional analysis of problem behavior: A quantitative and qualitative review. *Journal of Applied Behavior Analysis*.
<https://doi.org/10.1002/jaba.583>

Smith, R. G., & Churchill, R. M. (2002). Identification of environmental determinants of behavior disorders through functional analysis of precursor behaviors. *Journal of Applied Behavior Analysis*. <https://doi.org/10.1901/jaba.2002.35-125>

Thomason-Sassi, J. L., Iwata, B. A., Neidert, P. L., & Roscoe, E. M. (2011). Response latency as an index of response strength during functional analyses of problem behavior. *Journal of Applied Behavior Analysis*. <https://doi.org/10.1901/jaba.2011.44-51>

Free published resources for synthesized FA (PFA / IISCA). The open-ended interview, IISCA design form, data sheet, and graph template are free from the source: Hanley's Practical Functional Assessment site, practicalfunctionalassessment.com/implementation-materials. They aren't reproduced here; please get them there.

Assess, don't guess.



Thanks for doing this work, and for bringing a process to the table instead of a guess.

If you want help building this into a system across your whole caseload, that's what the **School BCBA Systems Transformation Program** is for:

behaviorschool.com/transformation-program

More from me: robspain.com

“I don't have a magical quick answer, but I have a process.”

— Robert Spain, BCBA, IBA